Full Stack Development Assignment

Multilingual Translation Web Application

March 13, 2026

Scenario

A startup company is developing a multilingual communication platform that helps users translate short messages between different languages in real time.

Your team has been tasked with building a **web-based translation application** that allows users to enter text, select languages, and receive translated results instantly.

This assignment simulates a real product feature that could be integrated into messaging platforms, travel applications, or international collaboration tools.

As a **Full Stack Developer**, you are responsible for implementing the frontend interface and integrating it with a translation API.

Objective

Develop an interactive **Translation Web Application** that allows users to translate text between languages using an external API.

Design Mockup

Implement the interface based on the provided design.

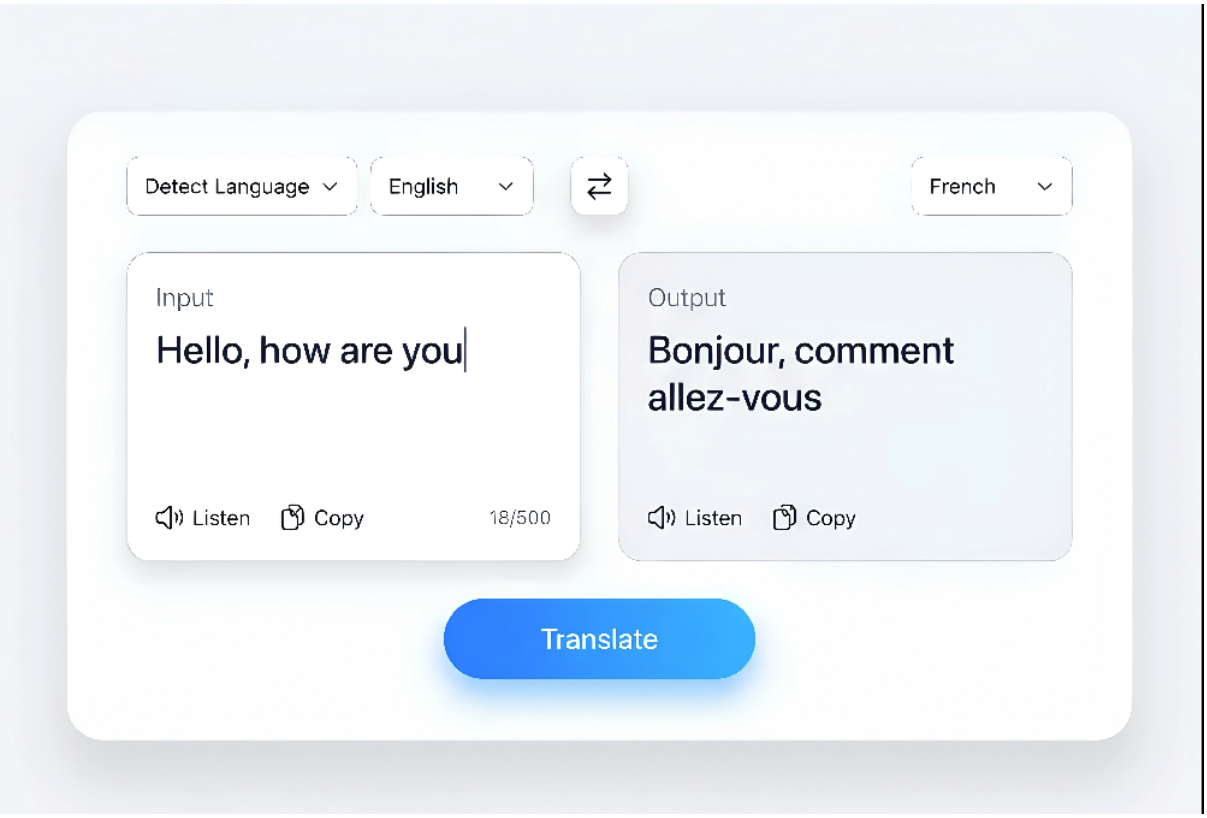

Figure 1:

Functional Requirements

Your application must include the following features:

1. The page should display a translation interface similar to the provided design.
2. By default, the application should translate the sentence:

   "Hello, how are you"

   from **English to French** when the page loads.
3. Users must be able to input their own text for translation.
   - Maximum character limit: **500 characters**
4. A **Translate** button should trigger the translation process.
5. (Optional Enhancement) Implement **real-time translation updates** using:
   - Debounce, or
   - Throttling
6. Users must be able to choose the source language from at least the following options:
   - Detect Language

    - English
    - French

7. Users must be able to choose the **target language** for translation.
8. The application should include a **switch language button** that swaps:
    - Source language
    - Target language
9. Users must be able to **listen to the text** using Text-to-Speech:
    - Listen to the original text
    - Listen to the translated text
10. Users must be able to **copy text** easily:
    - Copy the input text
    - Copy the translated text

Technical Requirements

The translation data must be retrieved using the following API:

`https://api.mymemory.translated.net/get`

Example request:

```
fetch("https://api.mymemory.translated.net/get", {
  method: "POST",
  body: JSON.stringify({
    q: "Hello, how are you",
    langpair: "en|fr"
  }),
  headers: {
    "Content-Type": "application/json"
  }
})
.then(response => response.json())
.then(data => {
  // Handle translated text
})
.catch(error => {
  // Handle error
});
```

Technology Stack

You may use any of the following technologies:

- HTML, CSS, JavaScript

Bonus Features (Optional)

Students who implement the following may receive additional credit:

- Real-time translation with debounce
- Loading indicator while translating
- Error handling for failed API requests
- Responsive mobile design
- Dark mode

Deliverables

Students must submit:

- Source code (GitHub repository link)
- README with setup instructions

Evaluation Criteria

Interface implementation	20%
API integration	25%
Application functionality	25%
Code structure and readability	15%
User experience features	15%

Submission Deadline

16/03/2026